

Name: Maths Class:

SYDNEY TECHNICAL HIGH SCHOOL



YEAR 12 HSC COURSE

Extension 2 Mathematics

Assessment 1 March 2010

TIME ALLOWED: 70 minutes

Instructions:

- Write your name and class at the top of this page, and on all your answer sheets.
- Hand in your answers attached to the rear of this question sheet.
- All necessary working must be shown. Marks may not be awarded for careless or badly arranged work.
- Marks indicated within each question are a guide only and may be varied at the time of marking
- START ALL QUESTIONS ON A NEW PAGE
- Approved calculators may be used.

(FOR MARKERS USE ONLY)

1	2	3	TOTAL
/17	/17	/16	/50

QUESTION 1: (17 Marks)

Marks

- 5 (a) If $z = 1 - \sqrt{3}i$, find
- (i) $\bar{z}$ (ii) $|z|$ (iii) $\arg z$ (iv) $\arg iz$ (v) $\frac{1}{z}$ (in simplest form)

- (b) Given the ellipse $9x^2 + 16y^2 = 144$, find

- 1 (i) the length of the major axis
- 1 (ii) the eccentricity
- 1 (iii) the co-ordinates of the foci
- 1 (iv) the equations of the directrices
- 1 (v) the slope of the tangent at the point $P(3, \frac{3\sqrt{7}}{4})$
- 1 (vi) the equation of the normal at $P(3, \frac{3\sqrt{7}}{4})$
(DO NOT SIMPLIFY THIS)

- 2 (c) (i) Sketch the region where the inequalities

$$|z - 2| \leq |z - 2i| \quad \text{and} \quad |z - 1 - 2i| \leq 1$$

hold simultaneously.

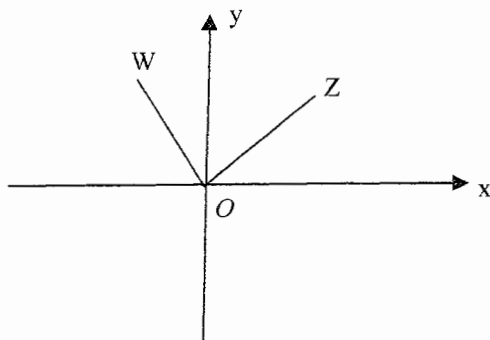
- 4 (ii) P is a point on the boundary of the region in part (i) above, and is represented by the complex number z , where $\arg z = \frac{\pi}{4}$.

Find the 2 possibilities for z (in the form $a+ib$).

QUESTION 2: (17 Marks)

Marks

- (a) The point Z, represents the complex number $z = 2 + 3i$



The line OZ is rotated anticlockwise by $\frac{\pi}{2}$ radians to form the line OW .

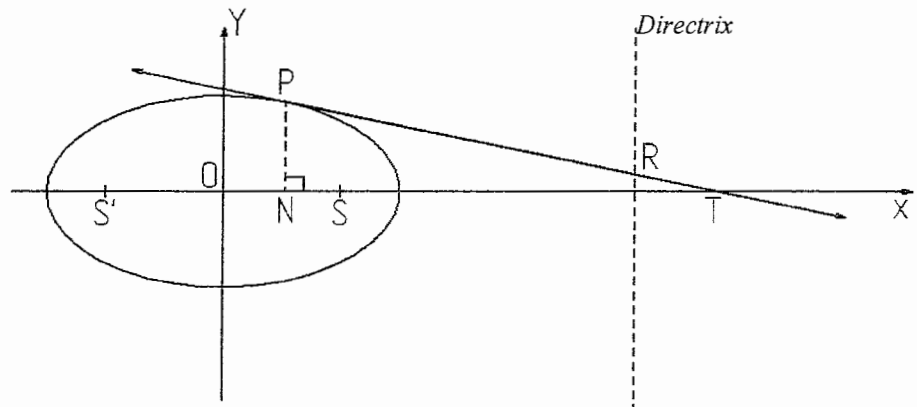
- 2 (i) Find the complex number w , represented by the point W.
- 2 (ii) Give the exact value of $\arg\left(\frac{z}{w}\right)$
- 3 (b) For any point Z, representing the complex number z , you are given that
- $$\arg(z - 1) - \arg(z - i) = 0$$
- On an Argand Diagram, draw the locus of the point Z.
- 5 (c) (i) Prove De Moivre's Theorem by the process of Mathematical Induction.
[NOTE: De Moivre's Theorem states that $(r \operatorname{cis} \theta)^n = r^n \operatorname{cis} n\theta$]
- 2 (ii) Express both $1 + i$ and $1 - i$ in the form $r \operatorname{cis} \theta$
- 3 (iii) Using De Moivre's Theorem, or otherwise, and your answers to part (b) above, find, as a whole number, the value of

$$(1 + i)^9 + (1 - i)^9$$

QUESTION 3: (16 Marks)

Marks

- (a) $P(x_1, y_1)$ is any point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$



The tangent at P cuts the major axis of the ellipse at T and the Directrix at R, while N is the foot of the perpendicular from P to the x-axis.

O is the centre of the ellipse, while S and S' are the foci.

- 4 (i) Show that the equation of the tangent at P is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$
(Show all working)
- 1 (ii) Find the co-ordinates of the point T.
- 1 (iii) Show that $ON \cdot OT = a^2$

QUESTION 3 continues overleaf....

.....**QUESTION 3 continued**

(b) You are given the curve $y^2 = x^2(4 - x^2)$,

1 (i) Find the points where this curve cuts the x-axis.

3 (ii) Use implicit differentiation, or otherwise, to show that $\frac{dy}{dx} = \frac{4x-2x^3}{y}$

(iii) By taking the positive square root of the curve only, the curve becomes

$$y = x\sqrt{(4 - x^2)}$$

1 Show that, in this instance, $\frac{dy}{dx} = \frac{2(2-x^2)}{\sqrt{4-x^2}}$

3 (iv) Hence find the co-ordinates of the turning points on the new curve $y = x\sqrt{(4 - x^2)}$ and identify their nature.

DO NOT ATTEMPT TO FIND THE SECOND DERIVATIVE

2 (v) Hence neatly sketch the original curve $y^2 = x^2(4 - x^2)$, showing all features found in the parts above.

SOLUTIONS

Teacher's Name:

Student's Name/N°:

QUESTION 1:

(a) $z = 1 - \sqrt{3}i$

(i) $\bar{z} = 1 + \sqrt{3}i$ (ii) $|z| = \sqrt{1+3} = 2$

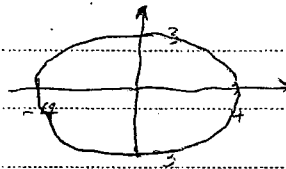
(iii) $\arg z = \tan^{-1}\left(\frac{-\sqrt{3}}{1}\right) = -\frac{\pi}{3}$ (iv) $-\frac{\pi}{3} + \frac{\pi}{2} = \frac{\pi}{6}$

(v) $\frac{1}{z} = \frac{1+\sqrt{3}i}{4}$

1 MARK EACH

(b) $\frac{x^2}{16} + \frac{y^2}{9} = 1$

(i) MAJOR AXIS is 8



1 MARK each part

(ii) $b^2 = a^2(1-e^2)$

$9 = 16(1-e^2)$

$e^2 = 1 - 9/16 \Rightarrow e = \sqrt{7}/4$

(iii) Foci are $(\pm\sqrt{7}, 0)$

(iv) Directrices are $x = \pm \frac{16\sqrt{7}}{7}$ (OR $16/\sqrt{7}$)

(v) $\frac{2x}{16} + \frac{2y}{9} \frac{dy}{dx} = 0$

$\therefore \frac{dy}{dx} = -\frac{2x}{16} \times \frac{9}{2y} = -\frac{9x}{16y}$

At $x=3, y = \frac{3\sqrt{7}}{4}, m_T = -\frac{9\sqrt{7}}{28}$

(vi) $m_N = \frac{28}{9\sqrt{7}}$

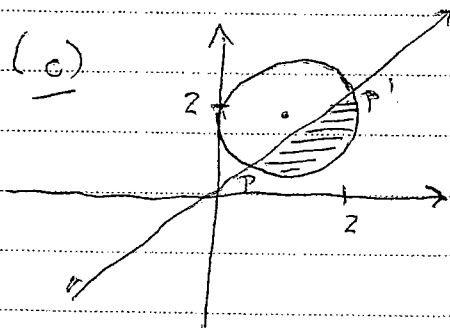
$y - \frac{3\sqrt{7}}{4} = \frac{28}{9\sqrt{7}}(x-3)$

$9y\sqrt{7} - \frac{27}{4} \times 7 = 28x - 84$

$112x - 36y\sqrt{7} - 147 = 0$

any of these forms for a mark

(c)



1 MARK for circle

1 MARK for the line

Teacher's Name:

Student's Name/N°:

$$\begin{aligned} \text{(ii)} \quad y &= x & (1) \\ (x-1)^2 + (y-2)^2 &= 1 & (2) \end{aligned}$$

$$x^2 - 2x + 1 + y^2 - 4y + 4 = 1$$

$$\therefore 2x^2 - 6x + 4 = 0$$

$$x^2 - 3x + 2 = 0$$

$$(x-2)(x-1) = 0$$

$$x = 2 \quad \text{OR} \quad x = 1$$

$$y = 2 \quad \text{OR} \quad y = 1$$

$\therefore P$ can be $(2, 2)$ or $(1, 1)$

and z can be $2+2i$ or $1+i$

① for these 2 equations

① mark for solving simultaneously

① mark each answer
= 2

SOLUTIONS

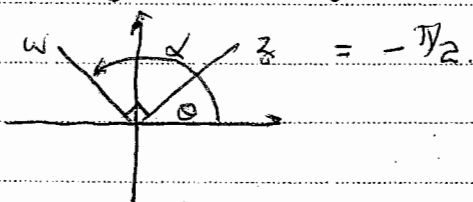
Teacher's Name:

Student's Name/N^o:

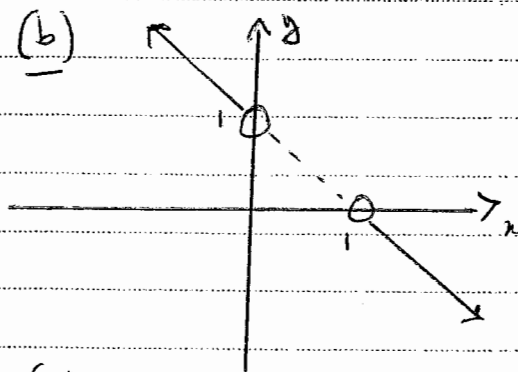
QUESTION 2:

(a) (i) $w = iz$
 $= \begin{cases} 2i - 3 \\ -3 + 2i \end{cases}$

(ii) $\arg(z/w) = \arg z - \arg w$



OR by rationalising the fraction and finding the answer was purely negative complex



(c) For $n=1$, $r \cos \theta = r \cos \theta$ trivial

(i) For $n=2$ $(r \cos \theta)^2 = r^2 (\cos^2 \theta - \sin^2 \theta + 2i \sin \theta \cos \theta)$
 $= r^2 (\cos 2\theta + i \sin 2\theta)$

$\therefore$ true for $n=1, 2$

Assume the formula is true for $n=k$

$\therefore (r \cos \theta)^k = r^k \cos k\theta$

For $n=k+1$

$(r \cos \theta)^{k+1} = (r \cos \theta)^k (r \cos \theta)$
 $= r^k \cos k\theta \cdot r \cos \theta$

$= r^{k+1} [\cos k\theta \cos \theta - \sin k\theta \sin \theta + i (\cos k\theta \sin \theta + \sin k\theta \cos \theta)]$

$= r^{k+1} [\cos(k+1)\theta + i \sin(k+1)\theta]$

$= r^{k+1} \cos(k+1)\theta$

$\therefore$ If the formula is true for $n=k$, it is true for $n=k+1$

BUT it is true for $n=2$, so it is true for $n=3$
 etc.

MARKS and

COMMENTS

1 for -thin
 1 for either } 2

2 MARKS

1 MARK for the line
 1 " for gap in mind
 1 " for open circles

① for testing 1,2

← ①

↓ ①

← ①

① for acceptable conclusion

Teacher's Name:

Student's Name/N^o:

QUESTION 2 CONT...

$$\begin{aligned} \text{(ii)} \quad 1+i &= \sqrt{2} \operatorname{cis} \pi/4 \\ 1-i &= \sqrt{2} \operatorname{cis} (-\pi/4) \text{ or } \sqrt{2} \operatorname{cis} (7\pi/4) \end{aligned} \quad \left. \vphantom{\begin{aligned} 1+i &= \sqrt{2} \operatorname{cis} \pi/4 \\ 1-i &= \sqrt{2} \operatorname{cis} (-\pi/4) \end{aligned}} \right\} \textcircled{1} \text{ each part}$$

$$\begin{aligned} \text{(iii)} \quad (1+i)^9 &= (\sqrt{2})^9 \operatorname{cis} 9\pi/4 \\ (1-i)^9 &= (\sqrt{2})^9 \operatorname{cis} (-9\pi/4) \end{aligned} \quad \left. \vphantom{\begin{aligned} (1+i)^9 &= (\sqrt{2})^9 \operatorname{cis} 9\pi/4 \\ (1-i)^9 &= (\sqrt{2})^9 \operatorname{cis} (-9\pi/4) \end{aligned}} \right\} \textcircled{1}$$

$$\therefore (1+i)^9 + (1-i)^9 = (\sqrt{2})^9 [\cos \pi/4 + i \sin \pi/4 + \cos(-\pi/4) + i \sin(-\pi/4)]$$

$$= (\sqrt{2})^9 [2 \cos \pi/4] \quad \leftarrow \textcircled{1}$$

$$= 16\sqrt{2} \times 2/\sqrt{2}$$

$$= 32 \quad \leftarrow \textcircled{1}$$

SOLUTIONS

Teacher's Name:

Student's Name/Nº:

QUESTION 3:

(a)(i) $2x/a^2 + \frac{2y}{b^2} \frac{dy}{dx} = 0$

$$\frac{dy}{dx} = -\frac{2x}{a^2} \times \frac{b^2}{2y}$$

$$= -\frac{xb^2}{ay}$$

① for this

At (x_1, y_1) $m_T = -\frac{x_1 b^2}{y_1 a^2}$

Tangent is

$$y - y_1 = -\frac{x_1 b^2}{y_1 a^2} (x - x_1)$$

$$yy_1 a^2 - y_1^2 a^2 = -xx_1 b^2 + x_1^2 b^2$$

①

$$\therefore yy_1 a^2 + xx_1 b^2 = x_1^2 b^2 + y_1^2 a^2$$

Divide by $a^2 b^2$

$$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2}$$

} ①

$$= 1$$

(ii) T is $(a^2/x, 0)$

1 MARK

(iii) $ON \cdot OT = x_1 \cdot \frac{a^2}{x_1}$

$$= a^2$$

1 MARK

(b)(i) $x = 2$ or $x = -2$

1 MARK

(ii) $2y \frac{dy}{dx} = (4 - x^2)2x + x^2(-2x)$

$$= 8x - 4x^3$$

} 2 MARKS

$$\frac{dy}{dx} = \frac{8x - 4x^3}{2y}$$

$$= \frac{4x - 2x^3}{y}$$

} 1 for simplifying

(iii) Using $y = x\sqrt{4-x^2}$

$$\frac{dy}{dx} = \frac{4x - 2x^3}{x\sqrt{4-x^2}}$$

1 MARK

$$= \frac{4 - 2x^2}{\sqrt{4-x^2}}$$

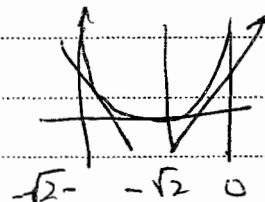
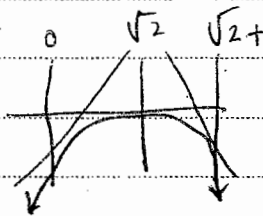
Teacher's Name:

Student's Name/N^o:Q 3 CONT...(iv) T.O.P.'s at $\frac{dy}{dx} = 0$

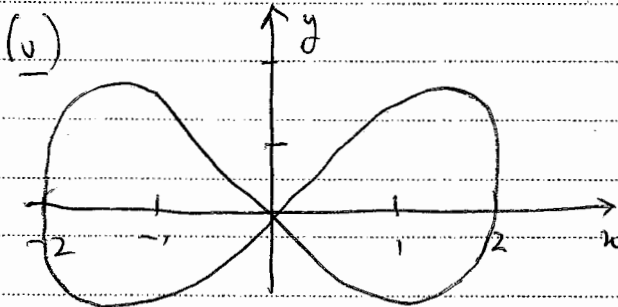
$$\therefore 2(2-x^2) = 0$$

$$\therefore \left. \begin{matrix} x = \sqrt{2} \\ y = 2 \end{matrix} \right\} \text{ or } \left. \begin{matrix} x = -\sqrt{2} \\ y = -2 \end{matrix} \right\}$$

① mark for each part = 2.

TESTING $\frac{dy}{dx}$ 

① mark

 $\therefore \text{max at } (\sqrt{2}, 2) \quad \text{min } (-\sqrt{2}, -2)$


② marks

only ① if only half a graph.